First-Year Roadmap: Java + DSA Foundation (Oct 2026 – Apr 2027)

Drafted Oct 19, 2026 · Window: 6 months, Oct 19, 2026 to about Apr 19, 2027

The goal in one paragraph

By Apr 2027: solid Java, about 150 DSA problems he can re-solve, 2 projects on GitHub, and clear spoken explanations. Google is the long-term target; this is the foundation phase.

Assumptions (please correct these)

- About 2 hours on most weekdays (sometimes more) and 5 to 6 hours on each weekend day, so roughly 15 to 18 hours a week.
- He is in sem 1 and has done no DSA yet. His Java level is "some", so Week 1 includes a quick check (below) to decide whether Month 1 can be shortened.
- Exam dates are unknown. In exam weeks he drops to "light mode": 30 to 45 minutes a day of revising old problems, nothing new.

Weekly routine

- [ ] Mon to Fri: 1 hr learn + 45 min problems + 15 min aptitude
- [ ] Sat: 3 hr DSA (1 pattern, 3 to 4 problems) + 2 hr Java or project
- [ ] Sun: re-solve last week's problems + explain 2 aloud + plan next week
- [ ] Rule: try 25 to 30 min alone, then hint, then re-solve after 7 days
- [ ] Exam weeks: light mode (30 to 45 min revision only)

Month 1 (Oct 19 to Nov 19): Java basics

- [ ] Week 1 check: FizzBuzz, reverse an array, class with constructor
- [ ] Create GitHub account, push every program
- [ ] Java: syntax, loops, methods, arrays, strings
- [ ] Java: OOP (classes, inheritance, interfaces)

- [ ] Big-O basics
- [ ] 15 easy array problems (NeetCode 150, in Java)
- [ ] Checkpoint: about 15 problems

Month 2 (Nov 19 to Dec 19): Collections, arrays, hashing

- [ ] Java: exceptions, ArrayList, HashMap, HashSet, StringBuilder, file I/O
- [ ] Finish the Abdul Bari course
- [ ] DSA: Arrays & Hashing, Strings (30 more problems)
- [ ] Git week: Learn Git Branching + Pro Git ch. 1 to 3
- [ ] Start Project 1 (Java console app)
- [ ] Checkpoint: about 45 problems, Java course done

Month 3 (Dec 19 to Jan 19): Two pointers, sliding window, binary search

- [ ] DSA: two pointers, sliding window, prefix sums, binary search (35 more)
- [ ] Finish Project 1 + README on GitHub
- [ ] Shell basics (1 week)
- [ ] One practice LeetCode contest (virtual, no pressure)
- [ ] Checkpoint: about 80 problems

Month 4 (Jan 19 to Feb 19): Stack, queue, linked list

- [ ] DSA: stack, queue, linked list, recursion intro (25 more)
- [ ] First live LeetCode weekly contest (aim Q1, try Q2)
- [ ] HTML + CSS on Scrimba (2 weeks)
- [ ] Checkpoint: 100+ problems

Month 5 (Feb 19 to Mar 19): Trees and recursion

- [ ] DSA: tree traversals, DFS/BFS intro, simple backtracking (25 more)
- [ ] Project 2: portfolio site on GitHub Pages
- [ ] One live contest every 2 weeks

- [ ] Checkpoint: about 130 problems, Project 2 live

Month 6 (Mar 19 to Apr 19): Consolidate

- [ ] Re-solve 30 old problems cold
- [ ] 2 mock interviews with you (45 min, talk aloud)
- [ ] One-page resume v1
- [ ] Plan next 6 months (DP, graphs, OS, DBMS)
- [ ] Checkpoint: about 150 problems, 2 projects, 6+ contests

Must-have tracks (alongside)

- [ ] Communication: 2-min spoken explanation after each problem
- [ ] Communication: record 1 explanation a week, listen back
- [ ] Communication: 1 short written note per DSA pattern
- [ ] Aptitude: 15 min, 4 days a week (IndiaBix)
- [ ] Logic: 1 puzzle a week on paper
- [ ] CS fundamentals: keep up with the college syllabus (maths, logic, programming)

Resources (only these)

- Java: Abdul Bari (Udemy) + MOOC.fi Java Programming
- DSA: NeetCode 150 only
- Git: Learn Git Branching + Pro Git ch. 1 to 3
- Shell: MIT Missing Semester
- Web: Scrimba HTML/CSS
- Aptitude: IndiaBix
- Parked until Month 6: the other 19 sheets, CodeChef, CS50

Open items for you

- [ ] Sem-1 exam dates
- [ ] Confirm weekday and weekend hours
- [ ] Result of the Week 1 Java check